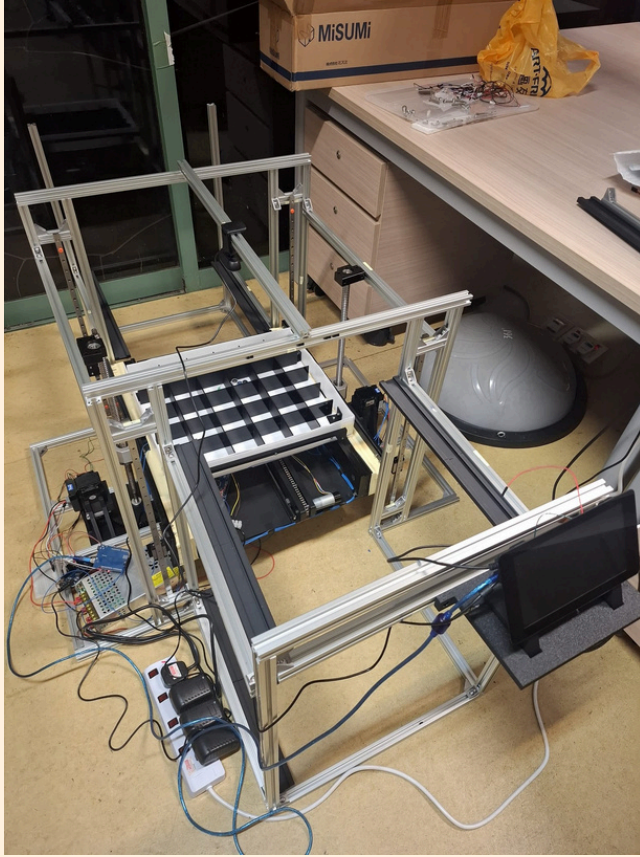
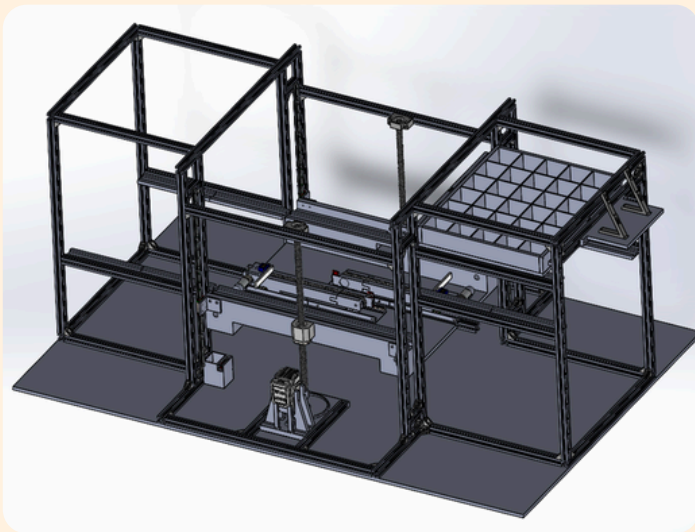


Shin Changsop's Portfolio

Design and Fabrication of a miniature Automatic Storage and Retrieval System (ASRS)



Prototype of mini-VLM



CAD model of mini-VLM

Problem Statement:

- Existing storage methods for small components are disorganised, with no reliable system for tracking component availability and location.
- The solution is to create a miniature version of the existing industrial Vertical Lift Module (VLM) with an embedded vision system and Webapp.

Solution Methodology:

- 2 Decoupled linear 1-DOF systems for raising and storing the trays, with their motors controlled by and intercommunicating with 2 Arduino Megas and 1 Raspberry Pi 5
- **Custom telescopic extractors** designed with **SolidWorks CAD** and fabricated through **Bambu P1S 3D printing**
- Developed a **custom web application, using Python Flask**, for intuitive tray selection and automated extraction control.
- Integration of a **YOLO vision system** to automate the tracking of components placed into the trays

Results:

- Fabrication of a working prototype that can undergo a full tray cycle in **82 seconds**
- Developed a compact mini-VLM prototype with a total volume of 1.046 m³, occupying **only 3.85% of the volume** of a small industrial VLM.